Support Vector Machines

Advanced Conceptual and Industry-Based MCQs

50 Questions

Instructions: Select the single best answer for each question. Base your reasoning on the stated business objective, data characteristics, and deployment constraints.

1. A legal-services company must classify documents into "relevant" and "irrelevant." It has 3,000 labelled documents represented by 80,000 sparse TF–IDF features. Its initial priority is a reliable, computationally manageable baseline. Which choice is most defensible?

(A) An RBF SVM because a larger feature count always requires a nonlinear kernel.

(B) A linear SVM because sparse, high-dimensional text can often be separated effectively without a nonlinear kernel.

(C) A one-class SVM because every document belongs to one of two classes.

(D) A hard-margin SVM because text features cannot contain label noise.

2. A factory has 600 labelled component images. A pretrained vision model already produces useful embeddings. The factory wants to compare classification approaches before collecting more data. Which approach best tests whether the existing representation is sufficient?

(A) Train a large vision network from scratch using only the 600 images.

(B) Assign labels using embedding magnitude alone.

(C) Use a one-class SVM while ignoring the available defect labels.

(D) Evaluate a regularized SVM on the embeddings using a suitable held-out evaluation.

3. An SVM-based fraud detector obtains 99.7% accuracy. Only 0.3% of transactions are fraudulent, and the detector predicts every transaction as legitimate. What is the most important conclusion?

(A) Accuracy conceals failure to identify the minority class; fraud recall and precision must also be examined.

(B) The model is suitable because accuracy exceeds 99%.

(C) The model must have learned an excellent separating margin.

(D) Increasing the number of legitimate transactions will necessarily resolve the problem.

4. A manufacturer penalizes missed defects more heavily than unnecessary inspections. It increases the SVM penalty for violations involving defective components. Which expectation is justified?

(A) All future defects will necessarily be detected.

(B) False positives will necessarily decrease.

(C) Training becomes more sensitive to violations involving defects, but operational trade-offs still require validation.

(D) The SVM decision score becomes a calibrated defect probability.

5. A sensor classifier uses temperature, vibration, and pressure. Pressure values are numerically thousands of times larger than the other measurements. An RBF SVM performs poorly. Which explanation should be investigated first?

(A) RBF kernels cannot process physical measurements.

(B) Unscaled features can cause pressure to dominate the distance calculations.

(C) Every feature must use the same physical unit.

(D) An RBF SVM requires all features to be binary.

6. A production team retrains an SVM but accidentally uses a new scaler when serving predictions from the old model. What is the principal risk?

(A) The model automatically adjusts its support vectors.

(B) The kernel becomes linear regardless of its original type.

(C) Predictions remain identical because scaling affects only training speed.

(D) Inputs are represented differently from training, potentially changing scores and predictions.

7. A machine-monitoring dataset contains thousands of overlapping time windows from each operating cycle. Randomly splitting windows produces excellent validation scores. Performance collapses on new cycles. Which issue is most likely?

(A) Correlated windows from the same cycle appeared in both training and validation.

(B) SVMs cannot classify time-window features.

(C) Validation requires every window to have a unique label.

(D) The number of windows was too large for any classifier.

8. A hospital develops an SVM using several recordings from each patient. It wants to predict outcomes for previously unseen patients. Which evaluation design best matches this goal?

(A) Randomly split individual recordings across folds.

(B) Keep each patient's recordings together when constructing training and validation partitions.

(C) Put each patient's longest recording in training and shortest recording in validation.

(D) Duplicate rare patients before splitting recordings randomly.

9. A bank wants to evaluate how an SVM trained today would perform on future transactions. Fraud strategies change over time. Which evaluation is most relevant?

(A) Evaluate on randomly selected training observations.

(B) Select a random split that gives the highest accuracy.

(C) Train on earlier transactions and evaluate on later ones, while preventing future information from entering training.

(D) Train on future transactions and evaluate on earlier ones.

10. A quality-control dataset contains both genuine difficult examples and some incorrect labels. A nearly hard-margin configuration performs poorly on new data. What is the strongest conceptual explanation?

(A) Allowing margin violations can prevent the classifier from fitting unreliable observations too aggressively.

(B) Maximizing a margin necessarily causes overfitting.

(C) Incorrect labels affect only probabilistic classifiers.

(D) Setting a larger penalty always improves generalization.

11. An RBF SVM achieves almost perfect training accuracy but poor validation accuracy. Both its penalty parameter and kernel locality parameter are large. Which interpretation is most plausible?

(A) The model is necessarily too simple.

(B) Feature scaling can no longer affect its predictions.

(C) Perfect training accuracy proves that the labels are correct.

(D) Strong pressure to fit training points combined with highly local similarities can produce an overly specific boundary.

12. An RBF SVM predicts almost the same class throughout a dataset, including on training examples. The kernel locality parameter is extremely small. What should the team consider?

(A) Every observation has necessarily become an outlier.

(B) Kernel similarities may vary too little to express the useful distinctions in the data.

(C) The model must have memorized individual observations.

(D) The training labels have become irrelevant to optimization.

13. A semiconductor company compares linear and RBF SVMs on engineered measurements. Their validation performance is nearly identical, but the linear model is much faster. Which decision is most defensible?

(A) Choose the RBF model because nonlinear models are always more accurate after deployment.

(B) Choose the model with more support vectors regardless of latency.

(C) Prefer the linear model if it meets the operational requirements and the performance difference is negligible.

(D) Reject both models because equivalent performance indicates an invalid experiment.

14. A network-security team has millions of labelled flows. An exact kernel SVM becomes expensive to train. Which strategy most directly addresses this limitation while retaining the possibility of nonlinear features?

(A) Use an approximate kernel feature map with a scalable linear classifier.

(B) Set the penalty parameter to infinity.

(C) Add more duplicate observations.

(D) Store the full kernel matrix at higher numerical precision.

15. Two RBF SVMs have similar validation performance. One uses 400 support vectors; the other uses 40,000. The system must score transactions under a strict latency limit. Which additional consideration matters most?

(A) The larger model necessarily has better calibration.

(B) The smaller model necessarily has lower training error.

(C) Support-vector count determines the number of classes.

(D) More support vectors generally increase kernel prediction work and model storage.

16. An embedded device must classify vibration features with limited memory. A linear SVM was trained on a large dataset. Why can its deployment representation remain compact?

(A) It must retain every training observation to classify new measurements.

(B) Its prediction function can be stored as a weight vector and intercept.

(C) It can discard its learned weights after training.

(D) Its predictions depend only on the number of training points.

17. A procurement team asks why a linear SVM flagged an invoice. The inputs are named, interpretable features. Which explanation is most appropriate?

(A) A feature with a large coefficient necessarily causes fraud.

(B) The support vector nearest the invoice is always its sole explanation.

(C) Feature contributions to the decision score can be examined, with attention to scaling and correlated inputs.

(D) Linear SVM decisions cannot be examined because all SVMs operate in infinite-dimensional spaces.

18. Two strongly correlated measurements receive very different linear SVM coefficients. A manager concludes that the larger coefficient identifies the only useful measurement. Why is this conclusion weak?

(A) Correlated features can share predictive information, and individual coefficients may depend on scaling and regularization.

(B) SVM coefficients are unrelated to predictions.

(C) Correlated inputs cannot be used by a linear SVM.

(D) The smaller coefficient proves that its feature was never observed during training.

19. An RBF SVM separates two product categories using laboratory measurements. A new observation lies extremely far from all training observations. The classifier gives a label. What should the team infer?

(A) A predicted label proves the observation resembles the training distribution.

(B) The observation must belong to the class with more training examples.

(C) The SVM has necessarily discovered a new category.

(D) The binary label alone does not establish reliable behavior outside the training distribution.

20. A customer-service system treats an SVM score of 0.8 as an 80% probability that a complaint requires escalation. What is the conceptual error?

(A) An SVM cannot produce positive scores.

(B) A raw decision score is not automatically a calibrated class probability.

(C) Every SVM score must lie between -1 and 1.

(D) Only negative scores require calibration.

21. An insurer needs reliable risk probabilities rather than only class labels. It plans to calibrate an SVM. Which design is most defensible?

(A) Fit calibration to the same in-sample scores used to optimize the SVM and assume reliability.

(B) Divide each score by the largest observed score.

(C) Fit calibration using held-out or out-of-fold scores and evaluate probability quality separately.

(D) Interpret the signed distance from the boundary as a probability without further modelling.

22. A fraud-review team can inspect only 200 transactions per day. Its SVM outputs continuous decision scores. Which operational approach best matches this constraint?

(A) Rank transactions and assess how many genuine frauds appear within the daily review budget.

(B) Require the default decision threshold regardless of the number of alerts.

(C) Report only overall accuracy.

(D) Randomly discard alerts until exactly 200 remain, without examining their scores.

23. A defect detector's decision threshold is lowered to send more items for inspection. With scores held fixed, what is the expected trade-off?

(A) Both true-positive and false-positive counts must decrease.

(B) Precision must improve.

(C) The SVM training margin must increase.

(D) More positives are predicted, so recall cannot decrease while false positives may increase.

24. A company compares two SVMs using independently chosen decision thresholds. One has greater recall but sends three times as many cases for manual review. What comparison is most useful?

(A) Compare recall alone because review cost is unrelated to classification.

(B) Compare performance at a common review budget or another explicitly defined operating constraint.

(C) Select the model with the larger absolute scores.

(D) Select the model with the higher training accuracy.

25. A one-class SVM is trained mainly on normal machine operation. It rejects a new operating mode as anomalous, although engineers consider that mode safe. What is the best interpretation?

(A) Every rejected observation is a mechanical fault.

(B) The rejection proves that the engineering assessment is incorrect.

(C) Novelty relative to the training data is not the same as operational danger.

(D) One-class SVMs cannot reject normal observations.

26. A factory has abundant normal data but almost no examples of every possible failure type. Why might a one-class SVM be considered?

(A) It can model a region associated with normal observations and flag departures without requiring labels for all failure types.

(B) It guarantees identification of the physical cause of every failure.

(C) It converts normal observations into labelled examples of every possible fault.

(D) It guarantees no false alarms when normal data are abundant.

27. A one-class SVM is trained on "normal" data containing a recurring undetected fault pattern. What risk does this create?

(A) Contamination always improves anomaly separation.

(B) The fault pattern must receive the most negative score.

(C) One-class training automatically verifies the labels.

(D) The model may incorporate the recurring fault into its learned notion of normal operation.

28. A machine has distinct normal operating regimes: idle, startup, and full load. A pooled anomaly detector creates frequent alarms during startup. Which investigation is most appropriate?

(A) Remove all startup observations from evaluation.

(B) Check regime coverage and consider conditioning the detector on operating mode.

(C) Increase the alarm threshold until no observation is ever rejected.

(D) Relabel every startup alarm as a genuine fault.

29. A team increases the ν parameter of a one-class SVM and claims that this specifies the exact future anomaly rate. What is the flaw?

(A) ν specifies the number of physical fault mechanisms.

(B) ν is a calibrated anomaly probability for every input.

(C) Its training-related bounds do not determine the exact anomaly rate on future data.

(D) ν has no role in the optimization problem.

30. A chemical plant uses support vector regression to estimate a continuously measured concentration. Small errors within the sensor's measurement tolerance are operationally unimportant. Why is ε-SVR conceptually suitable?

(A) Its loss can ignore residuals within a specified tolerance band.

(B) It converts concentration estimation into binary classification.

(C) It guarantees every prediction lies within sensor tolerance.

(D) It requires the target concentration to be normally distributed.

31. A forecasting team widens the ε-insensitive tube in SVR while keeping other settings fixed. What is the most defensible expectation?

(A) Every training observation must become a support vector.

(B) Every training residual must become zero.

(C) Predictions must become unbiased.

(D) More residuals may fall within the unpenalized region, potentially producing a sparser model.

32. A maintenance team uses SVR for remaining-useful-life prediction. It obtains low random-split error but poor performance on previously unseen machines. Which evaluation issue should be examined?

(A) Regression targets cannot be used with SVM methods.

(B) Measurements from the same machine may have appeared in both training and validation.

(C) Low random-split error guarantees reliable machine transfer.

(D) SVR requires every machine to have exactly the same lifespan.

33. A lender trains an SVM on loan applications that include a variable recorded only after the loan outcome is known. Why is excellent validation performance suspicious?

(A) Financial data are unsuitable for margin-based methods.

(B) SVMs cannot use binary input variables.

(C) The model may be using information unavailable at the actual decision time.

(D) Post-outcome variables are safe whenever cross-validation is performed.

34. A factory duplicates rare defective examples before randomly dividing the data into training and validation. Why can the resulting evaluation be misleading?

(A) Copies of the same original observation can appear on both sides of the split.

(B) Duplication necessarily changes every defect into a normal observation.

(C) Validation sets must contain no minority-class observations.

(D) An SVM cannot process two identical feature vectors.

35. A team selects features using labels from the complete dataset and then cross-validates an SVM on those features. Which change is needed for a sound evaluation?

(A) Perform feature selection only after deployment.

(B) Use training accuracy instead of cross-validation.

(C) Select the same number of features for every model and ignore how they were selected.

(D) Fit label-informed feature selection separately within each training fold.

36. An organization searches hundreds of SVM configurations and chooses the best cross-validation score. Why should it reserve an independent final evaluation or use nested cross-validation?

(A) Cross-validation cannot evaluate SVMs.

(B) Choosing the best configuration can also select favourable noise in the validation estimates.

(C) Hyperparameter searches make kernels mathematically invalid.

(D) An independent evaluation guarantees identical future performance.

37. A speech-authentication SVM performs well in a quiet laboratory but poorly on telephone recordings. What is the strongest first hypothesis?

(A) SVMs cannot classify audio embeddings.

(B) The penalty parameter must equal the number of speakers.

(C) Recording conditions have changed the input distribution or the quality of the representation.

(D) Every telephone recording belongs to the negative class.

38. A defect classifier uses image embeddings from a pretrained network. A linear SVM performs poorly. Which conclusion is most justified?

(A) The representation, the boundary model, the labels, and the evaluation setup all warrant investigation.

(B) All SVM methods are unsuitable for this industry.

(C) An RBF kernel is guaranteed to solve the task.

(D) Collecting additional data cannot help.

39. A retailer uses integer codes 1, 2, and 3 for three unordered product categories as an RBF SVM input. What concern arises?

(A) RBF kernels require category names rather than numbers.

(B) Three categories cannot be used in binary classification.

(C) Integer encoding automatically produces calibrated scores.

(D) The encoding introduces distances and ordering that may not reflect the categories' meaning.

40. An industrial classifier has many missing sensor values. The team imputes missing values using averages computed from the entire dataset before cross-validation. What is the best correction?

(A) Replace the SVM with a hard-margin formulation.

(B) Fit the imputation procedure using each training fold and apply it to that fold's validation data.

(C) Delete every validation observation with any missing value regardless of deployment conditions.

(D) Treat missing values as evidence of the positive class.

41. A safety classifier is trained mostly on common operating conditions. Rare but important conditions are absent. It achieves high validation accuracy on similarly sampled data. What does this result fail to establish?

(A) Whether the model can classify any common observation.

(B) Whether the training objective was computed.

(C) Whether the model performs adequately in the omitted rare operating conditions.

(D) Whether a binary SVM can produce class labels.

42. A cybersecurity team wants to use SVM margin magnitude as a confidence indicator. Which qualification is essential?

(A) Large margins do not, by themselves, guarantee calibrated confidence or correctness under distribution shift.

(B) Margin magnitude is identical to class probability.

(C) Every observation far from the boundary is necessarily inside the training distribution.

(D) Margin magnitude proves that an attack is impossible.

43. A company considers comparing raw score magnitudes from independently trained SVMs for different departments. Why can this be misleading?

(A) Every department must have the same positive-class prevalence.

(B) Independent SVMs always produce identical score scales.

(C) Raw scores are meaningful only for training observations.

(D) Score scales can differ with representation, regularization, and model formulation.

44. A multiclass document classifier uses one-versus-rest SVMs. Several classifiers produce positive scores for the same document. What does this indicate?

(A) The document must have been incorrectly labelled.

(B) Independent binary classifiers can make overlapping positive decisions, so a multiclass decision rule is needed.

(C) One-versus-rest training is mathematically infeasible.

(D) The classifiers must all have identical support vectors.

45. A product-recognition system was trained to distinguish ten known product types. A completely new product appears. What is a limitation of ordinary closed-set multiclass SVM use?

(A) It automatically creates an eleventh class.

(B) It necessarily returns a zero score for every known class.

(C) It can assign the new product to a known class unless an appropriate rejection mechanism is included.

(D) It cannot classify any observation absent from training.

46. A bank deploys an SVM whose training data are balanced, but fraud becomes much rarer in production. Even if class-conditional detection rates remain unchanged, which operational quantity can change substantially?

(A) The proportion of fraud among flagged transactions.

(B) The stored number of model coefficients without retraining.

(C) The kernel type chosen during training.

(D) The feature dimension of every incoming transaction.

47. A production team monitors only the percentage of positive SVM predictions. That percentage remains stable while model performance deteriorates. Why is this possible?

(A) A stable positive-prediction rate guarantees stable recall.

(B) Prediction rates uniquely determine classification accuracy.

(C) An SVM cannot deteriorate without changing its weights.

(D) Errors and underlying data relationships can change while the overall prediction rate stays similar.

48. An SVM fraud detector is retrained using only transactions that investigators previously reviewed. Most unflagged transactions never receive verified labels. What concern deserves attention?

(A) Reviewed cases necessarily represent all transactions.

(B) Label availability depends on earlier selection decisions, which can bias retraining and evaluation.

(C) Verified labels make sampling procedures irrelevant.

(D) The model can infer every missing label with certainty.

49. A factory proposes automatically retraining an anomaly detector on all recent observations after many alarms appear. Why might this be a poor response?

(A) Retraining can never accommodate a genuine process change.

(B) Recent data are always less informative than historical data.

(C) Without investigation, genuine faults may be incorporated into the updated definition of normal operation.

(D) An anomaly detector cannot be retrained once deployed.

50. A team must choose between an SVM and another classifier. The SVM has slightly higher overall accuracy, but worse recall for costly failures, greater latency, and no validated probability calibration. Which decision principle is strongest?

(A) Select the model using task-specific error costs, operating constraints, and properly validated performance.

(B) Select the SVM because accuracy overrides all other criteria.

(C) Select whichever model has more adjustable hyperparameters.

(D) Select whichever model achieves the highest training accuracy.